

UPS systems to be disrupted by an open hardware design

In a recent blog post, open source developer Eric Raymond wrote that the UPS system falls in a painful product category. He suggested that this whole product category needs to be disrupted by an open hardware design that will address the many deficiencies of existing hardware.

As the uninterruptible power supply (UPS) market is due for open source disruption, Raymond opened up about the Upside Project, a work-in-progress on GitHub. The Upside Project is currently defining requirements and developing a specification for a "...high quality UPS system that can be built from off-the-shelf parts in any reasonably well-equipped maker space or home electronics shop," he added.

In the original post called, 'UPSs suck and need to be disrupted', Raymond mentioned a set of complaints about what's sold to consumer/SOHO users. UPSs lack the kind of sensor information that protects car batteries. Raymond wrote that he wanted the UPS system to provide text-based alarm messages (rather than flashing lamps), and to provide decent monitoring information over USB.

Raymond informed that Eric Baskin is leading the hardware engineering team for Upside, Jay Maynard is developing the firmware, and Jeremy Mitts is copy editing the documents. He also said that their final deliverable will be PCB designs, a full parts list, assembly instructions, and full manuals for the hardware and software.

The Upside Project is working towards building UPS systems with open hardware, and with smart charging to preserve the battery. Instead of lead-acid, LiFePO (lithium iron phosphate) batteries are being suggested. The open hardware UPS system should be able to deliver 300W for 15 minutes.

helps reduce both development and deployment costs, allowing for a more streamlined system architecture—for example, electronic control unit (ECU) consolidation in automotive applications.

Though open source hypervisor options are currently available, none share ACRN's vision of an open source hypervisor solution optimised for embedded and IoT products.

"ACRN's optimisation for resource-constrained devices, its focus on isolating safety-critical workloads and giving them priority, make the project applicable across many IoT use cases. We're pleased to welcome project ACRN and invite embedded developers to get involved in the new community," said Jim Zemlin, executive director of The Linux Foundation.

ACRN is open to incorporate input from the open source, embedded, and IoT developer communities. The project will encourage collaboration and code contributions to the project. As of now, ADLINK, Aptiv, Intel Corporation, LGE, and Neusoft Corporation have already joined the ACRN project. ACRN will have a Linux-based service OS and the ability to simultaneously run multiple types of guest operating systems, providing a powerful solution for workload consolidation.

Indian startup becomes gold member of FIWARE

Delhi based Mobilepedia Innovations Pvt Ltd has become a gold member of



FIWARE Foundation, a global open source shared service platform provider.

While Mobilepedia is the first non-European gold member for FIWARE, it has also become the first company in India to receive a contract from the World Bank to pilot the FIWARE platform for one of India's smart cities.

The FIWARE pilot was formally announced by the World Bank during the recently organised Smart Republic 2018 conference. Numerous cities showed interest in the FIWARE pilot. And now, the World Bank has announced that Bareilly in Uttar Pradesh will be the first Indian city to implement the pilot.

The platform will allow Bareilly city to showcase an integrated platform that can act as a backbone for a 'control and command centre' for the smart cities.

Mukul Mahajan, founder, Mobilepedia Innovations, said, "We are looking forward to working closely with the FIWARE Foundation team to implement the FIWARE-based smart city platform for Indian cities. This can be a catalyst for the adoption of FIWARE by Indian smart cities."

FIWARE has standard APIs for data management and exchange. It has automated processes across the entire value chain, offering easy plug-and-play integration with other solutions and services.

Speaking about how becoming a gold member will help Mobilepedia, Mahajan shared, "Being a gold member, we get access to knowhow about a well established open source platform, which can drastically reduce the cost and time for implementation of shared services in the cities."

The association with FIWARE will further strengthen Mobilepedia's smart city solutions, monitoring and evaluation solutions, mobile health (IoT based) solutions and citizen engagement solutions.

As a market-ready open source software, FIWARE combines components that enable the connection to IoT with context information management and Big Data services in the cloud. Mobilepedia is a startup that delivers cloud computing and open data solutions to clients in South Asia. It has designed, developed and deployed municipal governance services in seven provinces across Sri Lanka.